Year 9 Science Test 1: mark scheme Core paper

Biology (40%): variation, adaptation, natural selection, cells · Chemistry (20%): states of matter, atoms · Physics (40%) · 40 minutes · 40 marks

DRAFT Not ready to print for students: Part 3: Physics (16 marks) has not been added yet.

Part	Marks	Share of the paper	Time
Part 1: Biology	16	40%	16 minutes
Part 2: Chemistry	8	20%	8 minutes
Part 3: Physics (to be added)	16	40%	16 minutes

Give the marks shown. Accept any correct equivalent wording. Where a part says "all four correct = 2 marks", do not award half marks.

Part 1: Biology 16 marks · about 16 minutes

1 Environmental variation (3 marks)

Part	Answer	Marks
(a)	temperature: abiotic; predation: biotic; amount of light: abiotic; infectious disease: biotic. All four correct = 2 marks; two or three correct = 1 mark.	2
(b)	height	1

2 Inherited variation (2 marks)

Part	Answer	Marks
(a)	nucleus (1), fertilisation (1).	2

3 Adaptation and extinction (2 marks)

Part	Answer	Marks
(a)	nature reserve: an area where the habitat and the species living in it are protected; breeding programme: endangered animals are bred in captivity; gene bank: seeds, gametes or cells stored at a low temperature. All three correct = 2 marks; one correct = 1 mark.	2

4 Natural selection (2 marks)

Part	Answer	Marks
(a)	Correct order: a few moths had genes that made them black (1); soot made the tree trunks black (2); birds ate more pale moths (3); each generation had more black moths (4). All four correct = 2 marks; first and last correct = 1 mark.	2

5 Cells and the microscope (3 marks)

Part	Answer	Marks
(a)	A cell wall, B nucleus, C chloroplast, D vacuole. All four correct = 2 marks; two or three correct = 1 mark.	2
(b)	x40 (10 x 4)	1

6 Natural selection in the Salton Sea STRETCH AND CHALLENGE (4 marks)

Part	Answer	Marks
(a)	By chance, some tilapia had genes (inherited variation) that helped them cope with salt (1). As the water got saltier, more of these fish survived than the others (1). They reproduced and passed on their genes (1), so each generation had more salt-tolerant fish, until all of them could cope (1). Do not credit 'the fish changed so they could cope'.	4

Part 2: Chemistry 8 marks · about 8 minutes

7 States of matter (2 marks)

Part	Answer	Marks
(a)	A gas, B solid, C liquid. All three correct = 2 marks; one or two correct = 1 mark.	2

8 Atoms and the periodic table (4 marks)

Part	Answer	Marks
(a)	(i) 11 protons (1). (ii) 23 - 11 = 12 neutrons (1).	2
(b)	2,8,1 (the missing number is 1).	1
(c)	group 1 (one electron in the outer shell).	1

9 Elements in the same group STRETCH AND CHALLENGE (2 marks)

Part	Answer	Marks
(a)	(i) fluorine 2,7 and chlorine 2,8,7, both correct (1). (ii) Both have seven electrons in the outer shell, and the outer electrons decide how an element reacts (1).	2

Part 3: Physics 16 marks · about 16 minutes

Physics questions to be added: 16 marks, about 16 minutes. Planned as 12 marks of Core questions and one 4-mark Stretch and Challenge question, matching Part 1. The topics taught in physics are needed before these can be written.

Total: 40 marks